

KARNATAK LAW SOCIETY'S

**GOGTE INSTITUTE OF TECHNOLOGY UDYAMBAG, BELGAVI-
590008**

(An Autonomous Institution under Visvesvaraya Technological University Belagavi)

(APPROVED BY AICTE, NEWDELHI)



MINI PROJECT REPORT

TOPIC: SOLDIER HEALTH MONITORING SYSTEM

Submitted in the partial fulfilment for the academic requirement of

5th Semester B.E in

ELECTRICAL AND ELECTRONICS ENGINEERING

Submitted by

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CERTIFICATE

This is to certify that Mr Rohan Mirje(2GI23EE037) , Miss. Sanjana Sajjan(2GI23EE041), Miss Swati Saidapur(2GI23EE053) students of KLS Gogte College can be requirements of fifth semester of **Bachelor of Engineering in Electrical and Electronics** considered as a Bonafide work for partial fulfillment for the completion of academic of the Visvesvaraya Technological University, Belagavi during the year 2025 - 2026 .

It is certified that all corrections/suggestions indicated have been incorporated in the report. The project report has been approved as it satisfies the academic requirements prescribed for the said Degree.

Signature of the Guide

Signature of the HOD

ABSTRACT:

The Soldier Health Monitoring System is designed to ensure the safety and well-being of soldiers in critical environments. This system utilizes an STM32 microcontroller, integrated with a heartbeat sensor, LM35 temperature sensor, buzzer, and GPS module, to monitor key physiological parameters in real-time. The heartbeat sensor detects the soldier's pulse rate, while the LM35 sensor tracks body temperature, both of which are vital indicators of health. In case of abnormal readings, the system triggers a buzzer to alert the soldier or nearby personnel. The GPS module continuously monitors the soldier's location, enabling quick response in emergencies, such as a health issue or injury. All collected data is displayed on an LCD, providing immediate feedback to the soldier or commanding officers. Additionally, the system can be extended to include wireless communication for remote monitoring by healthcare teams, enhancing response time in critical situations. This health monitoring solution aims to improve safety, provide early warning in case of health anomalies, and assist in swift intervention, ultimately ensuring the well-being of soldiers during their operations.

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INTRODUCTION:

In modern warfare and military operations, the health and safety of soldiers are paramount to ensuring the success of missions and the well-being of personnel. Soldiers are often exposed to high-stress environments, extreme physical exertion, and hazardous conditions, which can lead to various health risks. Monitoring their health in real-time is crucial for preventing and addressing medical emergencies, ensuring that prompt action is taken in case of health-related issues such as heart rate abnormalities, overheating, or injury. The Soldier Health Monitoring System addresses this need by continuously tracking vital health parameters such as heart rate and body temperature. Utilizing an STM32 microcontroller, this system incorporates a heartbeat sensor, an LM35 temperature sensor, a buzzer for emergency alerts, and a GPS module to monitor the soldier's location. The system is designed to provide immediate feedback on the soldier's health status, with the LCD display showing real-time data for easy access by the soldier or supervisory personnel. In addition to providing critical health information, the system aims to offer safety through its alert mechanism, which triggers a buzzer in case of abnormal readings, thereby prompting swift medical intervention. The inclusion of GPS tracking enables the monitoring of the soldier's position in the event of an emergency, facilitating faster response times and enhancing situational awareness. This innovative solution is not only intended for use in military operations but could also have applications in any field where monitoring the health of individuals in remote or high-risk environments is critical.

OBJECTIVES:

Real-time Health Monitoring:

- To continuously monitor the soldier's heart rate and body temperature using sensors (heartbeat sensor and LM35).

Early Detection of Health Anomalies:

- To detect abnormal health readings, such as irregular heart rate or elevated body temperature, and trigger an alert using a buzzer.

Location Tracking for Emergency Response:

- To provide continuous GPS tracking of the soldier's location for quick response in case of emergencies or medical issues.

Wireless Communication (Optional Expansion):

- To potentially extend the system's functionality to allow remote health monitoring and data transmission to a centralized base or medical team, enhancing situational awareness.

Reliable and Portable Design:

- To design a lightweight and portable system that can be worn or carried by soldiers without hindering their mobility or performance.

User-friendly Interface:

- To provide an easy-to-read LCD interface displaying health data and alerts in a clear, concise manner for the soldier in active field conditions.

METHODOLOGY:

The Soldier Health Monitoring System is designed to provide real-time monitoring of the soldier's health parameters such as heart rate and body temperature, while simultaneously tracking their location for emergency response. The system uses an STM32 microcontroller as the central controller to integrate all components and process data efficiently.

Below is a detailed explanation of how the system works:

1. Heartbeat Sensor (Pulse Sensor) :

- The heartbeat sensor is used to monitor the soldier's heart rate in real time. The sensor detects the pulse from the soldier's body, typically via a fingertip or earlobe.
- The sensor sends the pulse rate data to the STM32 microcontroller, where it is processed and displayed on the LCD screen.
- If the heart rate exceeds or falls below predefined safe limits, the system activates the buzzer to alert the soldier or nearby personnel about the abnormal condition.

2. LM35 Temperature Sensor :

- The LM35 temperature sensor continuously measures the body temperature of the soldier. The sensor provides an analog voltage corresponding to the temperature, which is then converted to a temperature reading by the STM32 microcontroller.
- The measured temperature is displayed on the LCD screen.
- If the body temperature crosses a predefined threshold (indicating potential heatstroke or hypothermia), the buzzer is triggered to provide an alert.

3. GPS Module :

- The GPS module tracks the soldier's geographic location in real-time, providing latitude and longitude coordinates.
- This data is essential for emergency response in case of health emergencies or injury, as it allows rescuers to quickly locate the soldier.
- The system can send these coordinates to a remote monitoring station (if communication is integrated), or display them on the LCD for the soldier's immediate reference.

4. LCD Display :

- The LCD screen acts as the primary user interface for the system. It continuously displays real-time data, including:
 - o Heart rate (beats per minute)
 - o Body temperature (in Celsius or Fahrenheit)
 - o GPS location (latitude and longitude)
- This information is updated regularly to ensure the soldier or monitoring personnel have accurate, up-to-date health data.

5. Buzzer (Alert System) :

- The buzzer is used as an alert mechanism. If the system detects abnormal values, such as a heart rate or temperature outside the safe range, or if the soldier's condition requires immediate attention, the buzzer sounds to notify the soldier or personnel nearby.
- The buzzer can also be triggered in case of a system error or malfunction, ensuring that any issues are promptly addressed.

6. STM32 Microcontroller :

- The STM32 microcontroller serves as the brain of the system. It handles all the data processing, from reading sensor data (heartbeat, temperature, GPS) to triggering alerts (buzzer) and displaying information on the LCD.
- It also allows communication with external devices (such as a wireless communication module) for remote monitoring if integrated.

7. Power Supply :

- The system can be powered by a rechargeable battery or a portable power source, ensuring that the soldier has continuous health monitoring during operations.
- Power management ensures that the system operates efficiently without draining the battery excessively, especially during prolonged missions.

COMPONENTS:

Hardware and Software Requirements:

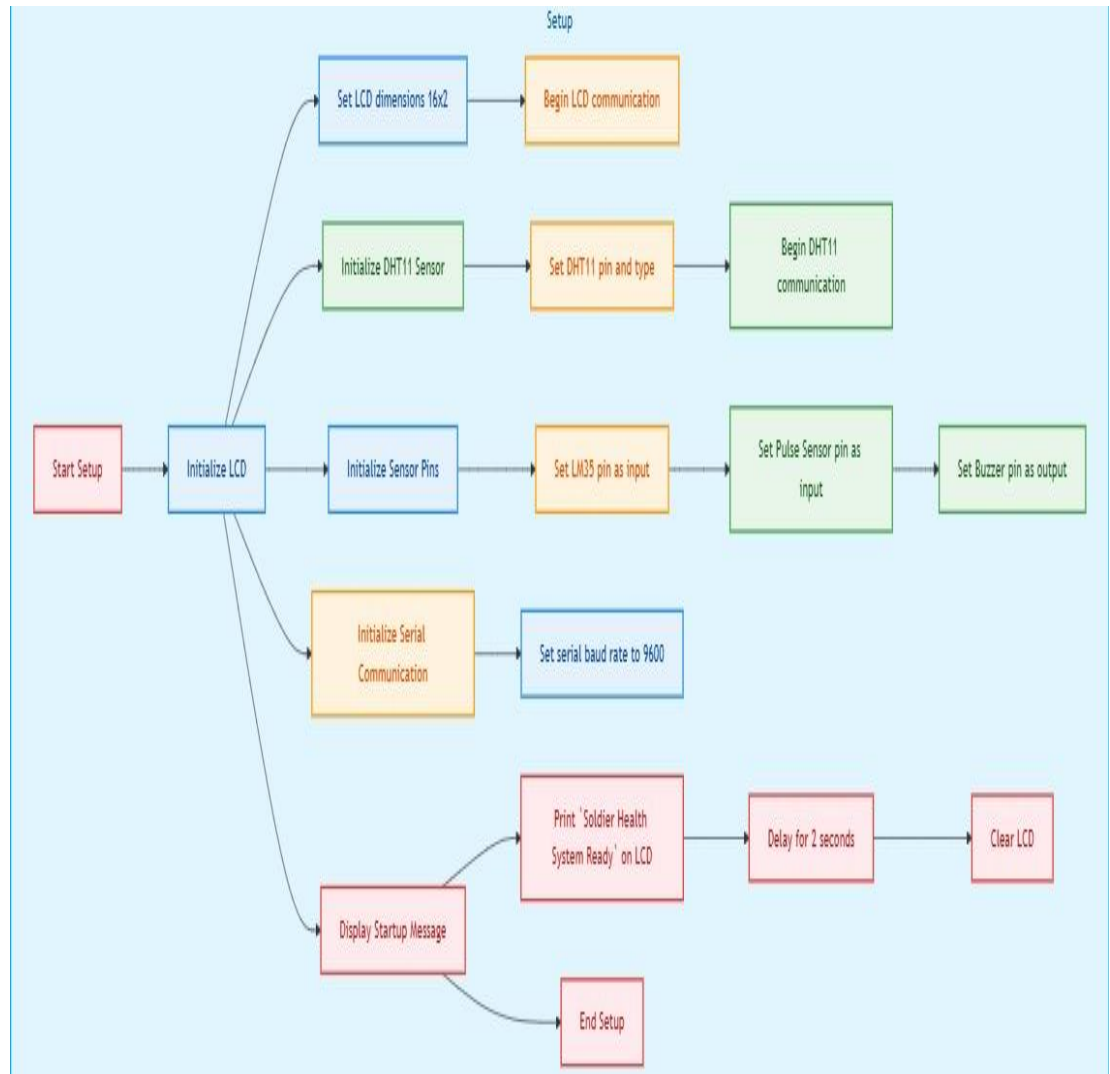
Hardware Requirements:

- STM32
- LCD Display
- Heartbeat Sensor
- Temperature Sensor
- Buzzer
- GPS
- Power Supply

Software Requirements:

- Arduino IDE
- Embedded C

FLOWCHART:



CODE :

```
#include <LiquidCrystal.h>
#include <DHT.h>

// ---- DHT11 ----

#define DHTPIN PB9
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);

// ---- LCD ----
LiquidCrystal lcd(PB13, PB12, PB0, PB1, PC13, PC14);

// ---- Sensors ----
#define LM35_PIN PA0
#define PULSE_PIN PA1
#define BUZZER PA8

void setup() {
  lcd.begin(16, 2);
  dht.begin();

  pinMode(LM35_PIN, INPUT);
  pinMode(PULSE_PIN, INPUT);
  pinMode(BUZZER, OUTPUT);

  Serial.begin(9600);

  lcd.print(" Soldier Health ");
  lcd.setCursor(0,1);
  lcd.print(" System Ready ");
  delay(2000);
  lcd.clear();
}

void loop() {
```

```

// ----- LM35 Temp -----
int raw = analogRead(LM35_PIN);
float voltage = raw * (3.3 / 4095.0);
float tempLM35 = voltage * 100.0;

// ----- DHT11 -----
float humidity = dht.readHumidity();
float tempDHT = dht.readTemperature();

// ----- Pulse Sensor -----
int pulse = analogRead(PULSE_PIN);

// ----- LCD Display -----
lcd.setCursor(0, 0);
lcd.print("T1:");
lcd.print(tempLM35);
lcd.print("C ");

lcd.print("H:");
lcd.print(humidity);

lcd.setCursor(0, 1);
lcd.print("T2:");
lcd.print(tempDHT);
lcd.print("C ");

lcd.print("P:");
lcd.print(pulse);

// ----- Alarm -----
if (pulse > 800 || tempLM35 > 38 || humidity > 80)
    digitalWrite(BUZZER, HIGH);
else
    digitalWrite(BUZZER, LOW);

delay(1000);
}

```

CONCLUSION:

The Soldier Health Monitoring System provides an effective solution for real-time tracking of vital parameters like heart rate, body temperature, and location. With the STM32 microcontroller, LCD display, and buzzer alerts, the system ensures quick detection of health issues and immediate response during critical missions. Its lightweight and portable design allows continuous monitoring without affecting a soldier's performance, ultimately improving safety and mission success.

Future Scope:

- Wireless remote monitoring through IoT or satellite communication for live updates to command centers.
- Integration of additional sensors such as blood oxygen (SpO₂), ECG, fatigue detection, or hydration level.
- AI-based health prediction to identify early signs of fatigue, stress, or medical emergencies.
- Data logging and cloud analytics for long-term health assessment and mission planning.
- Wearable form factor such as a smart vest or wristband for improved comfort and mobility.

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